

Criterion validity of heel-mounted inertial sensors for stride-length estimation during single- and dual-task walking in older adults with and without cognitive impairment

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ABSTRACT

Background. Gait metrics such as stride length are sensitive markers of fall and cognitive-decline risk in older adults, yet the reference systems that measure them accurately are limited to short walks in the laboratory. Wearable inertial sensors could extend monitoring into clinical and everyday settings, but their accuracy remains uncertain in the slow, variable gait of older adults.

Methods. We validated a heel-mounted inertial measurement unit (IMU) system against a GaitRite electronic walkway in 32 community-dwelling older adults (19 in the MoCA-defined cognitively normal group and 13 in the MoCA-defined cognitive impairment group). Participants completed single-task walking and three graded cognitive dual-task walking conditions (serial subtraction by 2s, 3s, and 7s) over a 6.1 m walkway. The pipeline combined bilateral stance detection with retroactive gravity-residual correction, used adaptive-threshold heel-strike and toe-off detection to support bilateral stance validation, and required no per-participant calibration.

Results. Across 10,472 matched strides retained after walk-level alignment quality control, stride length agreed with the walkway to within 3.9 cm root-mean-square error (RMSE), with a bias of +0.3 cm and an intraclass correlation (ICC) of 0.976. Mean absolute error fell below 4 cm in 31 of 32 participants. Per-condition RMSE ranged from 3.7 to 4.0 cm despite a 16% drop in gait speed under the most demanding condition. Gait speed, stride time, and cadence also agreed closely with the walkway (ICC 0.90–0.96). Participant-level error was modestly higher in the MoCA-defined cognitive impairment group than in the cognitively normal group. Replacing the internal mid-stance integration anchor with GaitRite heel-strike timing did not reduce stride-length RMSE.

Conclusions. A calibration-free heel-mounted IMU pipeline maintained low stride-length error across graded dual-task walking in older adults. Error was low in both MoCA-defined groups. These findings establish criterion validity for successfully synchronised, supervised straight walkway passes and support further clinical evaluation. Everyday-living validity, other clinical cohorts, and test-retest reliability remain untested.

1 INTRODUCTION

Gait metrics provide clinically relevant measures of mobility in older adults. Greater stride-to-stride variability is a strong predictor of incident falls in older adults (Maki, 1997; Verghese et al., 2009). Slower gait also predicts falls and cognitive decline (Verghese et al., 2009; Montero-Odasso et al., 2012). A meta-analysis reported that a baseline stride-length threshold of 0.64 m or less predicted adverse clinical events and mortality in older adults (Bytyçi and Henein, 2021). Instrumented walkways and optical motion capture provide criterion measurements during short, controlled laboratory walks (Webster et al., 2005; Arens et al., 2021). Wearable inertial measurement units (IMUs) can extend stride-length measurement beyond these settings, but the quality of these estimates depends on both the measurement setup and the processing method.

Sensor error, placement, and algorithm design all affect stride-length accuracy with foot-mounted IMUs. A common approach estimates stride length by integrating acceleration twice. Noise and small sensor errors accumulate during this process and cause the estimated position to drift (Wahlström and Skog, 2021). Sensor position and attachment influence the recorded signal and the resulting stride-length error (Küderle et al., 2022). Zero-velocity updates (ZUPTs) limit drift by correcting the velocity estimate during stance, when the foot is stationary (Foxlin, 2005; Wahlström and Skog, 2021). These stationary anchors can bound each integrated stride. Adaptive thresholds can improve zero-velocity detection across walking speeds compared with a single fixed threshold (Li et al., 2023).

Fixed amplitude thresholds can miss genuine stance phases and misclassify other stationary periods. During mid-stance the foot is still. Both the sensor's rotational speed (the gyroscope norm, the combined magnitude of triaxial angular velocity) and the moment-to-moment variation in its acceleration then drop close to zero. A threshold detector labels a sample as stance whenever both signals fall below fixed cutoffs (Skog et al., 2010). Fixed detector settings can fail when movement patterns differ from those used for tuning (Foxlin, 2005). Poor tuning can miss a true stance or label a non-walking stationary period as stance. A missed stance merges two adjacent integration intervals, while a false stance divides one interval.

Bilateral ZUPT checks each candidate stance against movement of the opposite foot. During normal walking, one foot lifts from the ground shortly after the opposite foot enters stance. Arens et al. used contralateral toe-off information to identify an ipsilateral ZUPT during post-stroke walking (Arens et al., 2021). Our detector applies the same biomechanical relationship by accepting an ipsilateral stance candidate when contralateral toe-off falls within the expected time window. A recovery step also searches for quiet periods near unmatched opposite-foot toe-offs, allowing the detector to recover stances that a threshold alone missed. This check targets missed and false stance detections during variable gait.

Two gaps remain in the evidence for this approach. The first is the lack of validation under cognitive load in older adults. One prior validation placed sensors on the foot and shank during straight, single-task walks in 10 healthy adults and 12 people with Parkinson's disease (Sijobert et al., 2015). Cognitive dual-task walking can slow gait and increase stride-to-stride variability (Beauchet et al., 2005; Yogev-Seligmann et al., 2008; Hausdorff et al., 2008). These changes broaden the range of gait speeds and patterns a pipeline must resolve and can make mid-stance harder to detect. The pipeline should be validated under these conditions rather than only during steady single-task walking. One heel-mounted IMU study assessed intra-session reliability in 101 middle-aged adults, including a cognitive dual-task condition, but did not validate against a criterion reference (Boutaayamou et al., 2025). Prior calibration-free validation included pathological cohorts walking at self-selected speed on an instrumented treadmill (Laidig et al., 2021). These protocols did not test graded cognitive load in older adults. Dual-task gait assessment has clinical relevance for evaluating fall risk and gait-cognition interactions (Montero-Odasso et al., 2012; Commandeur et al., 2018; Springer et al., 2006). To our knowledge, no prior study has validated gait-event detection and stride-length estimation against a criterion walkway in older adults under graded cognitive load across MoCA-defined cognitive status.

The second gap concerns heel-mounted accuracy in older adults. A multi-position study in 14 healthy young adults found higher single-stride error at the heel than at five other foot-mounted locations, particularly during fast walking (Küderle et al., 2022). Arens et al. used bilateral foot-mounted sensors and contralateral toe-off information in people post-stroke ($n = 8$), with optical motion capture as the

91 reference (Arens et al., 2021). A heel-mounted bilateral ZUPT stance-detection and stride-length pipeline
92 has not been validated against a criterion walkway in older adults walking under graded cognitive load.

93 **Our central contribution is validating two modifications to a calibration-free heel-mounted ZUPT**
94 **pipeline under graded cognitive load.** We validated the pipeline against a GaitRite electronic walkway
95 in 32 community-dwelling older adults. The cohort included 19 participants in the MoCA-defined
96 cognitively normal group and 13 in the MoCA-defined cognitive impairment group. Participants walked
97 under single-task and three graded cognitive dual-task conditions. The pipeline adds contralateral stance
98 validation and retroactive gravity-residual correction to standard per-stride ZUPT integration. It requires
99 no per-participant calibration. This design tests whether low stride-length error persists beyond steady
100 single-task walking.

101 2 MATERIALS & METHODS

102 2.1 Participants

103 **Thirty-two community-dwelling older adults completed the protocol after exclusions.** We recruited
104 33 community-dwelling older adults aged 60 years or older, of whom 32 contributed to the analysis. The
105 analysed cohort comprised 8 male and 24 female participants, with mean age 70.8 ± 7.8 years (range
106 60–83), mean height 167.4 ± 7.6 cm, and mean body mass 73.0 ± 13.2 kg (per-participant values in
107 Supplementary Table S1). Supplementary Table S2 reports the two participant-condition records whose
108 paired source data were unavailable. We formed two screening groups using the Montreal Cognitive
109 Assessment (MoCA) (Nasreddine et al., 2005). The groups included 19 cognitively normal participants
110 (CN, MoCA ≥ 26) and 13 participants meeting the MoCA criterion for cognitive impairment (CI, MoCA
111 14–25). These labels denote MoCA-defined screening groups rather than clinical diagnoses. Inclusion
112 criteria were a MoCA score of 14 or higher and the ability to walk 20 m independently without a mobility
113 aid. The University of Victoria Human Research Ethics Board approved the protocol (H22-00451), and
114 all participants provided written informed consent.

115 2.2 Instrumentation

116 **Heel-mounted IMUs and a pressure-sensitive walkway provided the data analysed in this study.** We
117 attached two wireless IMUs (Opal, APDM Inc., Portland, OR) to the posterolateral heel of each shoe,
118 adjacent to the calcaneus, using a 3D-printed housing fastened with a clip. We also placed an IMU over
119 the sacrum for the broader study protocol, but we did not use its data in this analysis. Prior validation
120 studies show the Opal sensor measures spatiotemporal gait parameters accurately against treadmill and
121 instrumented-walkway references in healthy adults and older adults (Washabaugh et al., 2017; Morris
122 et al., 2019). Each heel IMU recorded triaxial accelerometer (± 16 g) and gyroscope ($\pm 2000^\circ/\text{s}$) data at
123 256 Hz.

124 A pressure-sensitive electronic walkway (GaitRite, model Platinum, CIR Systems Inc., Franklin, NJ)
125 served as the criterion reference. The walkway has an active sensing area of $594 \text{ cm} \times 61 \text{ cm}$ with 1.27 cm
126 square sensors that record heel-on to toe-off stance events for each foot at 120 Hz. GaitRite gait parameters
127 show excellent concurrent validity against three-dimensional motion capture, with ICC 0.92–0.99 across
128 walking speed, cadence, step length, and step time (Webster et al., 2005). We synchronised the walkway
129 and IMU clocks using a GaitRite Pulse (GR Pulse) signal generated by a custom-written LabVIEW
130 program. The GR Pulse marked the start and end of each walkway pass on the LabVIEW clock. Within
131 each participant-condition pair, we fitted a linear clock correction between GR Pulse rising edges and
132 corresponding IMU activity onsets across up to seven walks. We then estimated a residual constant offset
133 within each walk by aligning adaptive-detector IMU heel strikes with GaitRite heel strikes.

134 2.3 Protocol

135 **Each participant completed four walking conditions, one single-task and three graded dual-task**
136 **conditions.** Single-task walking served as the baseline, followed by three dual-task conditions of graded
137 cognitive difficulty. Participants counted backwards aloud by 2s, 3s, and 7s from a random three-digit
138 starting number while walking. Each condition comprised seven trials of two passes (out and back) over
139 the GaitRite mat at a self-selected comfortable pace. This yielded 14 walks per condition and 56 walks
140 per participant across the four reported conditions. Participants walked beyond both ends of the mat to
141 allow acceleration and deceleration off the recording surface. Participants wore a safety belt, and a spotter
142 accompanied them. We provided rest periods between conditions as needed.

143 2.4 Signal preprocessing

144 **We bandpass-filtered the gyroscope signal and aligned the left-right sign convention before further**
145 **processing.** We first bandpass-filtered raw gyroscope signals (0.5–10 Hz, fourth-order Butterworth) to
146 remove low-frequency drift and high-frequency noise. We then inverted the left heel gyroscope Z-axis
147 to match the right heel sign convention (positive = plantarflexion angular velocity in the sagittal plane).
148 We left the triaxial accelerometer signals unfiltered for the ZUPT integration pipeline. The per-stride

149 zero-velocity and gravity-residual corrections addressed integration drift directly, so we did not add an
 150 accelerometer filter whose cutoff would require further tuning.

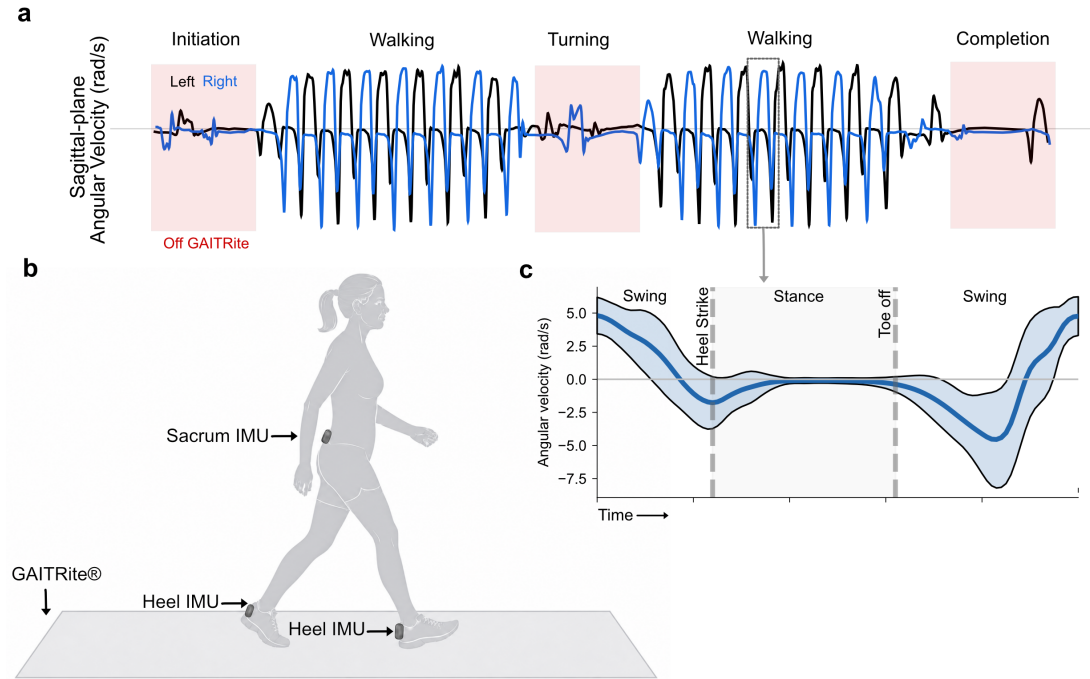


Figure 1. Experimental setup and gyroscope processing. (a) Filtered sagittal-plane signals from the left (black) and right (blue) heel IMUs during one out-and-back trial. Labels mark activity phases, red bars mark off-mat intervals, and the highlighted box is expanded in panel c. (b) Sensor placement during GaitRite walking. Heel IMUs were clipped to the posterolateral aspect of each shoe using 3D-printed housings. A sacrum-mounted IMU was present as part of the broader protocol, but its data were not used in this analysis. (c) Mean right-heel gyroscope Z-axis signal over a normalized gait cycle, pooled across participants and conditions. The Z-axis captures sagittal-plane angular velocity. The blue line shows the mean, shading shows ± 1 SD, dashed lines mark heel strike and toe off, and labels identify swing and stance.

151 2.5 Gait event detection

152 **We detected heel strikes and toe-offs on each foot using thresholds that adapted to the participant's**
 153 **walking pattern.** A fixed cutoff can miss events when stride amplitudes change with walking pace. We
 154 therefore adapted an established sagittal-plane gyroscope detector (Salarian et al., 2004) using signal-
 155 derived thresholds (Greene et al., 2010). The detector estimated stride frequency from the repeating
 156 pattern of the filtered gyroscope signal. Heel-strike and toe-off peak thresholds used 35% and 30% of
 157 the negative-peak amplitude interquartile range, respectively. Cadence scaled both thresholds. The peak
 158 threshold could not fall below 15% of the maximum absolute signal amplitude or 0.5 rad/s. Positive
 159 swing confirmation used 25% of the maximum absolute amplitude with a 1.0 rad/s floor. The detector
 160 then classified negative signal peaks according to the pattern that followed within 80–220 ms. A rapid
 161 rise into swing identified toe-off, while a near-zero segment identified heel strike (Fig. 1c). Toe-offs
 162 supported bilateral stance validation, and both events were compared with GaitRite for the timing analysis.
 163 Stride time for paired analyses came from consecutive stance midpoints because these stationary anchors
 164 also bounded each integrated displacement. We analysed discrete heel-strike timing separately against
 165 GaitRite initial contacts.

166 2.6 Ipsilateral stance phase detection with contralateral validation

167 **Stance-phase detection runs in two steps.** A low-motion detector first finds candidate stance windows
168 on one foot. A toe-off on the opposite foot then confirms each candidate. Here, ipsilateral means the foot
169 being evaluated and contralateral means the opposite foot. The resulting stance mask anchors the ZUPT
170 integration. We define stride boundaries at the midpoints of consecutive stance phases because the foot is
171 stationary there, providing a stable zero-velocity reference.

172 *Step 1. Ipsilateral low-motion detection.* The detector evaluated gyroscope magnitude and accelerometer
173 variability from the same heel sensor, following established fixed-threshold approaches (Skog et al.,
174 2010; Foxlin, 2005). We used a sliding 40 ms window. A sample qualified as a stance candidate when
175 both signals fell below adaptive thresholds calculated for each participant and walking condition. The
176 gyroscope criterion identified low rotation during foot-flat contact, while the accelerometer criterion
177 rejected brief low-rotation periods during swing. Algorithm 1 lists the thresholds, limits, and recovery
178 window used in the analysis.

179 *Step 2. Opposite-foot confirmation.* The detector used opposite-foot toe-off information to confirm stance
180 candidates, following the bilateral approach of Arens et al. (2021). We accepted a candidate when an
181 opposite-foot toe-off fell within 150 ms of either boundary of the candidate segment. Candidates without
182 a matching toe-off were rejected. An unmatched toe-off triggered a recovery search for a quiet period on
183 the other foot. The detector also removed very short stance segments, joined segments separated by brief
184 gaps, and checked unusually long strides for a missed stance. These steps reduced false detections during
185 stops and prevented two strides from being merged.

186 *Fallback to threshold-only detection.* The pipeline reverts to ipsilateral threshold-only stance detection
187 whenever contralateral validation returns an implausibly low stance count. We defined this as fewer than
188 three validated segments or fewer than 50% of the threshold-only segments when the threshold-only
189 detector found at least ten. This guard fired in 11 of 256 subject-condition-foot units (4.3%). Eight of the
190 11 units came from a single participant whose contralateral toe-off detection failed in every condition.
191 Most paired strides therefore used contralateral validation, while the fallback prevented loss of otherwise
192 usable data when contralateral event detection failed.

193 2.7 ZUPT stride length integration

194 **We estimated stride length by double-integrating the accelerometer signal between consecutive**
195 **stance midpoints.** During each stance phase the foot rests on the ground at zero velocity. The stance
196 mask from Section 2.6 flags these zero-velocity intervals, which are used to correct the drift that double
197 integration would otherwise produce (Foxlin, 2005; Skog et al., 2010). We estimated gyroscope and
198 accelerometer biases once per walking bout, defined as a run of consecutive stance phases separated by
199 no gap longer than 3 s. The stance phase supplies the reference, because true angular rate is zero and
200 true acceleration equals gravity. We bias-corrected the signals with these estimates before integration,
201 removing the slow sensor offsets that would otherwise accumulate as drift.

202 2.7.1 Orientation tracking

203 **We tracked sensor orientation with a Mahony attitude-and-heading reference system.** This filter
204 combines gyroscope and accelerometer measurements to estimate how the sensor is oriented relative
205 to gravity (Mahony et al., 2008). We used a higher correction gain during stance ($K_p = 1.0$), when the
206 stationary foot provides a reliable gravity reference, and a lower gain during swing ($K_p = 0.05$), when
207 movement affects the accelerometer signal. The integral gain remained at $K_i = 0.02$. We reduced the
208 proportional accelerometer correction gain during swing to limit contamination by dynamic acceleration.
209 This follows the movement-gating rationale of foot-mounted filters that disable accelerometer-based
210 orientation updates during swing (Hannink et al., 2017).

211 2.7.2 Gravity removal, double integration, and stride length extraction

212 **We integrated the gravity-removed acceleration twice between consecutive stance midpoints to**
213 **estimate stride length.** We rotated the bias-corrected accelerometer signal into the navigation frame

using the Mahony-estimated orientation, subtracted gravity, and double-integrated to obtain velocity and position (Foxlin, 2005). Because the foot is momentarily stationary at each stance boundary, its true velocity is zero. We applied a linear zero-velocity correction that distributes the accumulated velocity error backward across the preceding swing samples and then resets the velocity to that known zero, cancelling the drift built up over the stride. Each stride integrates independently with position and velocity reset at each stance midpoint, which bounds drift to a single swing phase. Stride length is the magnitude of the horizontal component of the 3D displacement between consecutive stance midpoints. We accepted strides with duration 0.4–2.0 s, length 0.3–2.5 m, and minimum gyroscope amplitude 3.0 rad/s, a low floor that excludes near-stationary segments rather than genuine strides.

2.7.3 Retroactive gravity-residual correction

We corrected the acceleration residual that remained after orientation tracking and gravity removal. The code refers to this step as bidirectional smoothing. Small orientation errors during swing leave a horizontal acceleration residual after gravity removal, which can bias the integrated stride length (Foxlin, 2005; Hannink et al., 2017). The correction reduced this positive offset in the current cohort (Supplementary Table S4).

We corrected this residual retroactively after each swing phase. At each stance onset we measured the residual. We skipped the first 10 ms to avoid the heel-strike impact transient, then averaged the navigation-frame acceleration over the next 20 ms. We then subtracted a linear ramp of this residual through the preceding swing samples, scaled by a gain of 0.5:

$$\mathbf{a}_{\text{corrected}}(t) = \mathbf{a}_{\text{nav}}(t) - 0.5 \cdot \frac{t}{T_{\text{swing}}} \cdot \mathbf{r} \quad (1)$$

where t is time from swing onset, T_{swing} is swing duration, and $\mathbf{r}$ is the gravity residual at the next stance onset. We selected the gain of 0.5 empirically in this cohort because no prior work specifies a value for this correction. This selection was not independent of the validation sample. The correction operates per stride and requires no external calibration. A sensitivity analysis across gains from 0.3 to 0.7 showed pooled RMSE values from 3.83 to 4.10 cm (Supplementary Table S7).

2.7.4 Algorithm summary

Algorithm 1 summarises the complete processing sequence. Step 1 builds the stance mask using low-motion detection and opposite-foot confirmation. Step 2 tracks orientation, removes gravity, integrates each stride, and applies the retroactive gravity-residual correction before resetting velocity. Contralateral stance validation and the gravity-residual correction distinguish the tested pipeline from a standard ZUPT implementation (Foxlin, 2005; Skog et al., 2010).

2.8 GaitRite stride matching

We paired each ZUPT stride with its GaitRite counterpart using clock alignment and stride-level checks. The GR Pulse correction placed each walk on the IMU timeline, and heel-strike alignment supplied a final offset within each walk. We accepted a walk-level alignment only when at least 60% of the expected GaitRite heel strikes had an IMU match within 150 ms. This required most reference events to match while allowing one or two missed detections in a short pass. Walks without a valid alignment were classified as synchronisation failures and did not enter paired stride or event-agreement analyses. We paired strides from the same foot when the IMU and GaitRite stride midpoints were each other's closest match. The allowable time difference adapted to stride duration and never exceeded 0.5 s. We retained GaitRite strides with lengths of 0.30–2.00 m and times of 0.40–1.80 s. The matched IMU length and time also had to fall between 50% and 150% of the corresponding reference value. We call this the stride-pair mismatch filter. It removed likely pairing errors and GaitRite records in which a missed heel strike produced an apparent double stride.

We retained only strides that fell completely within a GaitRite walk window. This removed turns, off-mat acceleration and deceleration, mat-entry transitions, and rest periods. Participants began walking approximately 1 m before the mat, so gait-initiation strides were not part of the synchronised comparison. Partial steps at either end of the mat were removed before matching.

Algorithm 1: Bilateral ZUPT stride-length estimation.

Input : Ipsilateral 3-axis accelerometer (*accel*) and gyroscope (*gyro*)
Contralateral toe-off times (*contra_to*)
Sampling rate (*fs*)

Output : Stride lengths (*stride_lengths*)

/ Step 1. Stance detection with contralateral validation */*

- 1 *gyro_norm* = magnitude of *gyro* at each sample
- 2 *accel_var* = 40 ms sliding variance of *accel*
- 3 *gyro_thresh* = clip(15th percentile of *gyro_norm*,
- 4 0.3, 2.0 rad/s)
- 5 *accel_thresh* = clip(25th percentile of *accel_var*,
- 6 0.2, 5.0 (m/s²)²)
- 7 *candidates* = samples where *gyro_norm* < *gyro_thresh* **and** *accel_var* < *accel_thresh*
- 8 **for** *each candidate segment* **do**
- 9 **if** *any contra_to event falls within ±150 ms of the segment boundary* **then**
- 10 accept segment as stance
- 11 **else**
- 12 reject segment
- 13 **end**
- 14 **end**
- 15 recover missed stances near unmatched *contra_to* using relaxed gyroscope-only threshold (25th percentile, 0.5 rad/s floor)
- 16 reject stance segments < 30 ms, merge segments separated by < 75 ms, split strides > 1.8 s at a mid-stride quiet period
- /* Step 2. Per-stride integration with retroactive gravity-residual correction */*
- 17 estimate *gyro_bias* and *accel_bias* from stance samples of each walking bout
- 18 **for** *each stride between consecutive stance midpoints* **do**
- 19 track orientation with Mahony AHRS (gain 1.0 during stance, 0.05 during swing)
- 20 rotate (*accel* − *accel_bias*) into the navigation frame, subtract gravity
- 21 measure gravity residual at the next stance onset
- 22 subtract a linear ramp of that residual through the preceding swing (gain 0.5)
- 23 integrate corrected acceleration → *velocity*, distribute the accumulated velocity error linearly back across the swing, then set *velocity*=0 at stance ends
- 24 integrate *velocity* → *position*
- 25 *stride_length* = horizontal distance between stride start and stride end positions
- 26 reject stride unless duration 0.4–2.0 s, length 0.3–2.5 m, peak *gyro_norm* > 3.0 rad/s
- 27 **end**
- 28 **return** *stride_lengths*

261 **The final analysis included 10,472 paired strides from 32 participants.** We excluded one of the 33
262 enrolled participants because the required heel-mounted IMU recordings were unavailable. We retained
263 every reported participant-condition record that had the required source data and at least one stride pair
264 after the common matching rules. We did not exclude records based on match count or stride-length error
265 because both are study outcomes. GR Pulse records were unavailable for ID_186 L2 and ID_195 L2, so
266 these two conditions could not enter the paired analysis (Supplementary Table S2). Of 1,764 available
267 walks, 1,669 passed alignment QC and 95 were classified as synchronisation failures. Failures occurred in
268 14 single-task, 13 serial-2, 19 serial-3, and 49 serial-7 walks. We also excluded three individual walks
269 for documented data-quality problems described in Supplementary Section S2. After stride-level checks,
270 the final pool contained 2,544 single-task, 2,618 serial-2, 2,770 serial-3, and 2,540 serial-7 strides. The
271 conditions included 32, 30, 32, and 32 participants, respectively.

2.9 Outcome measures and statistical analysis

The primary outcome was pooled stride-level RMSE between IMU and GaitRite stride length. Agreement outcomes included signed bias, SD of the error, 95% limits of agreement, MAE, ICC(2,1), and proportional bias. We calculated bias and limits of agreement following Bland and Altman (1986) and ICC(2,1) following Shrout and Fleiss (1979). We also reported the percentage of strides within 5 and 10 cm and calculated RMSE and MAE at the participant level. Event-detection outcomes were miss rate, timing bias, and RMSE against GaitRite heel strikes and toe-offs. A missed event had no IMU match within 150 ms. To assess stride-to-stride consistency, we calculated stride-length coefficient of variation for each participant and condition and compared the IMU and GaitRite values using bias, RMSE, and ICC(2,1).

We evaluated gait parameters other than stride length using RMSE, bias, and ICC(2,1). We also performed two design comparisons. First, we ran the full stride-length pipeline with bilateral and ipsilateral threshold-only stance detection while keeping all other parameters identical. We compared matched stride pairs, RMSE, bias, and ICC between methods. Second, we compared the internal mid-stance integration boundaries with GaitRite-derived heel-strike boundaries. For this comparison, we used RMSE, bias, MAE, SD of the error, 95% limits of agreement, and ICC(2,1).

We tested the effect of cognitive load on stride-length accuracy with a repeated-measures analysis of variance (ANOVA) on participant-level mean absolute error, restricted to the 30 participants with valid data in all four conditions. We applied the Greenhouse–Geisser correction when Mauchly’s test indicated a sphericity violation. We then ran pairwise condition comparisons with Bonferroni correction and focused reporting on each dual-task condition against single-task walking. We report generalised eta-squared (η_G^2) for the overall ANOVA. We compared participant-level MAE between the MoCA-defined groups with a two-sided Mann–Whitney test. This subgroup analysis was exploratory and was not adjusted for multiple comparisons. We described the stride-level relationship between gait speed and absolute error using linear regression and reported R^2 without using the regression for participant-level inference. Participant MAE confidence intervals in Fig. 3 used 2,000 walk-cluster resamples. We ran all analyses in Python 3.13 with pingouin 0.6.1, statsmodels 0.14.6, and scipy 1.17. We selected and reported reliability and agreement measures following the Guidelines for Reporting Reliability and Agreement Studies (GRRAS) (Kottner et al., 2011).

We assessed cohort-level accuracy with a participant-clustered bootstrap. We resampled the 32 participants with replacement and recomputed RMSE, MAE, bias, and ICC on the resulting stride pool. We repeated this process 10,000 times to obtain 95% percentile confidence intervals. The sample size was based on participant availability rather than a prospective power calculation.

We ran two sensitivity analyses. First, we repeated the accuracy analysis without the stride-pair mismatch filter defined above. All other matching and filtering rules remained unchanged. This analysis tested whether the filter affected the reported accuracy. Second, we varied the retroactive gravity-residual correction gain from 0.3 to 0.7 in increments of 0.1 while keeping the rest of the pipeline unchanged. We compared retained pairs, RMSE, MAE, bias, and ICC(2,1). These same-cohort analyses assess robustness to analysis choices but do not replace independent validation.

3 RESULTS

3.1 Gait event detection accuracy

The adaptive-threshold detector matched GaitRite heel strikes within 33 ms RMSE and toe-offs within 39 ms RMSE. Heel-strike miss rates ranged from 0.0% to 0.4% and toe-off miss rates from 2.8% to 3.8% across conditions (Table 1). Heel-strike RMSE was 27–33 ms and toe-off RMSE was 33–39 ms. A missed event denotes a heel strike or toe-off that the detector could not localise within a stride the pipeline had identified.

Table 1. Adaptive-threshold gait-event accuracy. HS = heel strike; TO = toe off. Miss rate is the percentage of GaitRite events unmatched within 150 ms. Bias = mean IMU-minus-GaitRite timing error.

Condition	Event	Miss rate (%)	<i>N</i> matched	Bias (ms)	RMSE (ms)	MAE (ms)
Single-task	HS	0.0	3749	−0.7	26.8	16.7
	TO	2.8	3698	+3.9	32.8	22.7
Serial 2s	HS	0.2	3650	−0.2	31.2	18.8
	TO	3.2	3586	−0.7	37.8	25.9
Serial 3s	HS	0.4	3858	+0.1	32.4	19.8
	TO	3.6	3760	0.0	38.7	26.6
Serial 7s	HS	0.4	3554	+0.1	32.6	20.1
	TO	3.8	3499	+1.7	39.3	27.3

3.2 Stance detection and event-timing comparisons

Bilateral validation recovered more reference strides with similar accuracy. On the GaitRite-labelled strides, bilateral validation recovered 10,472 matched pairs compared with 9,708 for threshold-only detection (Table 2). The two methods differed by 0.1 cm RMSE and both returned an ICC of 0.976.

Substituting GaitRite heel-strike timing for the internal mid-stance anchor did not reduce stride-length error. We re-ran the ZUPT integrator with GaitRite-derived heel-strike times as displacement endpoints on the 10,459 strides matched in both anchoring schemes. RMSE rose from 3.9 to 4.1 cm and ICC fell from 0.976 to 0.973 (Table 2).

Table 2. Pipeline design comparisons. The blocks compare stance detection and integration anchors on their respective shared stride sets, so *N* differs. All other pipeline steps were held constant. Bias = IMU minus GaitRite.

Variant	<i>N</i> pairs	RMSE (cm)	Bias (cm)	MAE (cm)	ICC(2,1)
<i>Stance detection</i>					
Bilateral (canonical)	10,472	3.9	+0.3	2.5	0.976
Ipsilateral (threshold-only)	9,708	3.8	−0.3	2.4	0.976
<i>Gait-event timing anchor</i>					
Mid-stance (canonical)	10,459	3.9	+0.3	2.5	0.976
GaitRite heel strike	10,459	4.1	+1.1	2.8	0.973

3.3 Stride length accuracy

The pipeline measured stride length with 3.9 cm RMSE and 2.5 cm MAE. The pooled analysis included 10,472 matched stride pairs and returned an ICC of 0.976. Bias was +0.3 cm and the SD of error was 3.9 cm (Table 3, Fig. 2). The scatter plot shows agreement across the full stride-length range (Fig. 2a). The Bland-Altman plot shows a small mean bias and a weak trend across the range (Fig. 2b; proportional-bias slope 0.036). Per-condition RMSE ranged from 3.7 to 4.0 cm, and per-condition ICC ranged from 0.971 to 0.977. The conditions included 32, 30, 32, and 32 participants (Table 3). The error fell within ± 5 cm for 87% of stride pairs and within ± 10 cm for 97%. Proportional bias was weak (slope

0.036, $R^2 = 0.026$). Relative to the mean GaitRite stride length of 1.20 m, pooled RMSE was 3.2% and MAE was 2.1%.

Bias was positive across all four conditions and error distributions overlapped (Table 3). Participant-level MAE ranged from 1.6 to 4.7 cm, with 31 of 32 participants below 4 cm (Fig. 3). Mean per-participant RMSE was 3.8 cm and ranged from 2.1 to 7.2 cm.

When we disabled the retroactive gravity-residual correction on the same matched strides, pooled RMSE and bias increased (Supplementary Table S4).

Table 3. Bilateral ZUPT stride-length accuracy by condition. n = participants; N = matched stride pairs; bias = mean IMU-minus-GaitRite error; LoA = Bland–Altman 95% limits of agreement.

Condition	n	N pairs	Bias (m)	RMSE (m)	MAE (m)	95% LoA (m)	ICC(2,1)
Single-task	32	2,544	+0.007	0.040	0.027	[−0.069, +0.084]	0.973
Serial 2s	30	2,618	+0.001	0.037	0.024	[−0.072, +0.073]	0.977
Serial 3s	32	2,770	+0.000	0.039	0.024	[−0.077, +0.077]	0.975
Serial 7s	32	2,540	+0.003	0.039	0.025	[−0.073, +0.078]	0.971
Pooled	32	10,472	+0.003	0.039	0.025	[−0.073, +0.078]	0.976

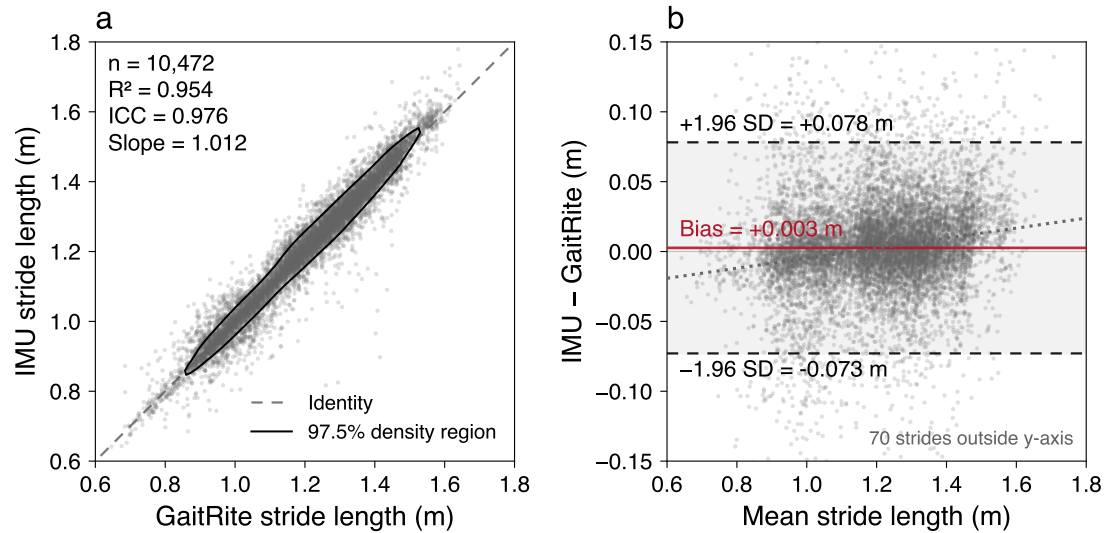


Figure 2. Stride length agreement between IMU and GaitRite. (a) All 10,472 matched strides. The dashed line marks identity, and the contour encloses the densest 97.5% of points. Linear fit $R^2 = 0.954$, slope = 1.012, and ICC(2,1) = 0.976. (b) Bland–Altman plot. The red line marks mean bias (+0.003 m), dashed lines mark the 95% limits of agreement (−0.073 to +0.078 m), and the dotted line shows proportional bias (slope = 0.036).

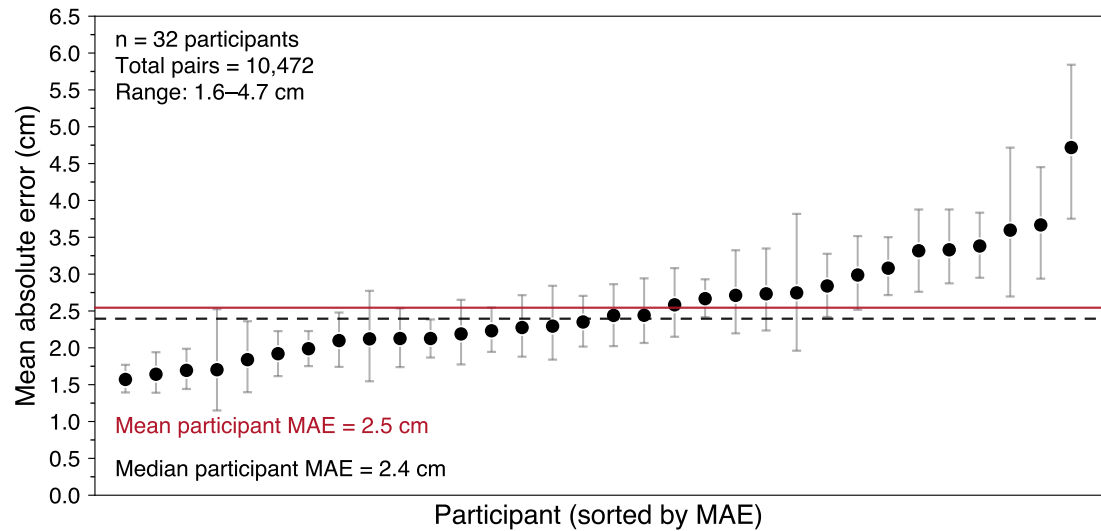


Figure 3. Participant stride-length accuracy across conditions ($n = 32$). Dots show participants sorted by MAE with 95% CIs. The solid red and dashed grey lines mark the mean (2.5 cm) and median (2.4 cm), respectively.

3.3.1 Effect of cognitive load and cognitive status on stride length accuracy

Accuracy did not differ across walking conditions, but participant-level error was modestly higher in the MoCA-defined CI group. The repeated-measures ANOVA on participant-level MAE included the 30 participants with data in all four conditions. It found no evidence of a condition effect ($F(2.35, 68.27) = 0.67$, Greenhouse–Geisser corrected $p = .541$, $\eta_G^2 = 0.016$). No comparison between single-task and a dual-task condition was significant after Bonferroni correction. Absolute error was also weakly related to gait speed ($R^2 = 0.002$). Mean signed bias was below 1 cm in every condition (Table 3).

Stratified by MoCA-defined cognitive status, RMSE was 3.65 cm in CN participants ($n = 19$, 6,139 strides) and 4.14 cm in CI participants ($n = 13$, 4,333 strides). Median per-participant MAE was 2.28 cm in the CN group and 2.84 cm in the CI group (Mann–Whitney $U = 71$, $p = .046$). Error remained low in both screening groups, although participant-level error was modestly higher in the CI group.

3.4 Robustness of accuracy estimates

Participant-clustered bootstrap confidence intervals show stable cohort-level accuracy under participant resampling. We resampled the 32 participants with replacement 10,000 times and recomputed accuracy metrics on each resampled cohort's stride pool. RMSE was 3.86 cm (95% CI [3.51, 4.23]) and MAE was 2.50 cm (95% CI [2.30, 2.70], Supplementary Table S5). Bias remained small (+0.26 cm, 95% CI [−0.08, +0.63]). ICC was 0.976 (95% CI [0.964, 0.981]).

Typical absolute error remained similar without the mismatch filter. Removing the stride-pair mismatch filter added 22 strides and increased RMSE from 3.86 to 5.49 cm, while MAE changed from 2.50 to 2.65 cm (Supplementary Table S6). The larger RMSE change reflects a small number of apparent missed or merged stride events. Across correction gains from 0.3 to 0.7, RMSE ranged from 3.83 to 4.10 cm and was 3.86 cm at the selected gain of 0.5 (Supplementary Table S7).

3.5 Other spatiotemporal gait parameters

The IMU reproduced the graded gait changes caused by cognitive load. GaitRite gait speed dropped 16.2% from single-task to serial 7s (1.19 to 0.99 m/s), stride time lengthened (1.09 to 1.22 s), and cadence fell (111 to 100 steps/min, Table 4). Mean GaitRite stride length also differed across conditions in the 30 participants with complete data ($F(1.39, 40.17) = 88.99$, Greenhouse–Geisser corrected $p < .001$). IMU

values tracked the reference values across all four conditions. Pooled ICC(2,1) was 0.961 for gait speed, 0.905 for stride time, and 0.902 for cadence.

Stride-length variability showed only moderate pooled agreement. Per-participant stride-length CV was 3.58% for the IMU and 3.08% for GaitRite. The pooled bias was +0.50 percentage points and ICC(2,1) was 0.576 (Table 4).

Table 4. Spatiotemporal agreement by condition across 10,472 matched strides. GaitRite and IMU values are mean (SD), and bias = IMU minus GaitRite. Stride-length CV is calculated within participant and reported as the cohort mean.

Parameter	Condition	GaitRite	IMU	Bias	RMSE	ICC(2,1)
Gait speed (m/s)	Single-task	1.186 (0.207)	1.189 (0.215)	+0.002	0.070	0.945
	Serial 2s	1.039 (0.231)	1.043 (0.237)	+0.004	0.064	0.963
	Serial 3s	0.992 (0.229)	0.994 (0.233)	+0.002	0.062	0.964
	Serial 7s	0.994 (0.196)	0.998 (0.202)	+0.004	0.063	0.949
	Pooled	1.051 (0.230)	1.054 (0.236)	+0.003	0.065	0.961
Stride time (s)	Single-task	1.088 (0.097)	1.093 (0.110)	+0.005	0.063	0.816
	Serial 2s	1.181 (0.160)	1.177 (0.158)	−0.004	0.071	0.900
	Serial 3s	1.209 (0.168)	1.207 (0.168)	−0.002	0.071	0.911
	Serial 7s	1.218 (0.165)	1.216 (0.165)	−0.002	0.073	0.903
	Pooled	1.175 (0.159)	1.174 (0.160)	−0.000	0.070	0.905
Cadence (steps/min)	Single-task	111.2 (9.9)	110.9 (11.1)	−0.33	6.18	0.827
	Serial 2s	103.3 (12.8)	103.7 (13.2)	+0.36	5.85	0.899
	Serial 3s	101.1 (13.1)	101.3 (13.5)	+0.21	5.57	0.912
	Serial 7s	100.2 (12.5)	100.4 (12.9)	+0.18	5.67	0.900
	Pooled	103.9 (12.9)	104.0 (13.4)	+0.11	5.82	0.902
Stride-length CV (%)	Single-task	2.69	3.24	+0.56	1.12	0.504
	Serial 2s	3.18	3.39	+0.21	0.82	0.827
	Serial 3s	3.34	3.89	+0.55	1.74	0.518
	Serial 7s	3.12	3.79	+0.66	1.60	0.489
	Pooled	3.08	3.58	+0.50	1.38	0.576

373 4 DISCUSSION

374 **This validation of bilateral heel-mounted IMUs across two MoCA-defined cognitive-status groups**
375 **yielded four main findings.** First, bilateral heel-mounted IMUs matched GaitRite stride length to within
376 3.9 cm pooled RMSE with ICC 0.976 across 10,472 paired strides in older adults walking under single-
377 and dual-task conditions. Second, bilateral validation recovered more GaitRite reference strides than
378 threshold-only detection with similar accuracy. Third, per-condition RMSE ranged from 3.7 to 4.0 cm
379 despite a 16% drop in gait speed. Fourth, error remained low in both MoCA-defined groups, although
380 participant-level error was modestly higher in the CI group.

381 **Our main advance is criterion validation of a practical heel-mounted system across graded cognitive**
382 **load in older adults.** Heel-mounted sensors remain uncommon in gait-validation studies (Lin et al.,
383 2025; Küderle et al., 2022). Previous studies reported low-centimetre errors at other foot and lower-leg
384 locations, but their cohorts, reference systems, and protocols differed from ours (Mariani et al., 2010;
385 Trojaniello et al., 2014; Laidig et al., 2021; Rampp et al., 2015). Direct performance rankings are therefore
386 inappropriate. Our study adds criterion validation in 32 older adults across single-task and graded dual-
387 task walking, with 10,472 paired strides and both RMSE and MAE reported. A multi-position study in
388 younger adults found higher single-stride error at the heel than at other foot locations, especially during
389 fast walking (Küderle et al., 2022). A later heel-mounted validation also found greater spread than a
390 sensor embedded in the shoe, although mean errors remained small (Luo et al., 2024). Embedded sensors
391 may be preferable when footwear modification is practical. We chose a clip-on heel attachment because it
392 requires no shoe modification and can be applied quickly in clinical or home settings.

393 **Accuracy remained stable across cognitive load, while cognitive-status results warrant further study.**
394 Per-condition RMSE stayed between 3.7 and 4.0 cm even as cognitive load slowed gait by 16%. A
395 separate validation used IMUs on both feet, sternum, and lower back in healthy and stroke adults. It found
396 only small changes in method disagreement under irregular stepping (Ensink et al., 2023). We found no
397 evidence of a condition effect on participant-level MAE ($F(2.35, 68.27) = 0.67$, Greenhouse–Geisser
398 corrected $p = .541$, $\eta_G^2 = 0.016$). The 0.3 cm spread across condition RMSE values was modest, although
399 the study was not designed as an equivalence test. Stride-level RMSE was 3.65 cm in CN participants
400 and 4.14 cm in CI participants. Participant-level MAE was modestly higher in the CI group ($p = .046$).
401 A sensitivity analysis restricted to the twelve CI participants with MoCA scores of 18–25 returned
402 4.01 cm RMSE (Supplementary Table S3). This subgroup analysis is exploratory and concerns MoCA
403 screening categories rather than clinical diagnoses. Diagnostic utility and longitudinal monitoring require
404 prospective studies with formal clinical classification and repeated measurements.

405 **The immediate application is supervised gait assessment when an instrumented walkway is im-**
406 **practical.** The heel clips require no footwear modification, and the pipeline adapts without participant
407 calibration. The stable error across task difficulty supports research and clinical evaluation during straight
408 walking passes. The present data do not support diagnostic use or unsupervised monitoring.

409 **The design comparisons supported the retroactive correction and internal mid-stance anchor.**
410 Disabling the retroactive gravity-residual correction increased bias from +0.3 to +2.2 cm and RMSE
411 from 3.9 to 4.9 cm (Supplementary Table S4). A small horizontal acceleration can remain after gravity
412 removal and accumulate when integrated across the stride. Correcting this residual reduced the offset
413 without a calibration walk or reference device. Heel-strike timing stayed inside the 150 ms validation
414 tolerance (Table 1). Replacing the internal mid-stance boundaries with GaitRite heel strikes did not
415 improve accuracy (Table 2). Bilateral validation recovered more matched strides than threshold-only
416 detection with similar accuracy. Prior foot-worn systems have included turning tasks (Mariani et al.,
417 2010). This bilateral detector’s performance during stops and turns remains untested.

418 **The pipeline measured mean gait parameters more accurately than stride-to-stride variability.** Gait
419 speed, stride time, and cadence produced pooled ICC values of 0.90–0.96 (Table 4), consistent with
420 prior IMU studies using sensors on the feet, ankles, and shanks (Trojaniello et al., 2014; Washabaugh
421 et al., 2017). In contrast, pooled stride-length CV agreement was moderate (ICC 0.58), and the IMU
422 overestimated CV by 0.50 percentage points. Accurate mean stride length therefore did not guarantee
423 equally accurate stride-to-stride variability. Variability estimates should be interpreted cautiously until
424 they are validated in longer walking bouts with more strides per condition.

425 **Five design constraints limit generalisation of our results.** First, GaitRite ground truth covers only
426 supervised, straight strides across a 6.1 m walkway. The protocol did not test turns, uneven surfaces,
427 stops, footwear variation, long continuous bouts, or unsupervised activity. It therefore cannot establish
428 everyday-living accuracy or validate the bilateral detector's intended rejection of non-walking events.
429 Those questions require a longer-duration criterion such as synchronised video or wearable-compatible
430 motion capture. Second, participants walked independently with relatively typical gait patterns and the
431 sample skewed towards female (24 of 32 participants). Generalisation to severe gait impairments and sex-
432 specific accuracy claims remains untested. Third, cognitive classification relied on MoCA screening rather
433 than formal neuropsychological assessment. The subgroup findings concern screening categories, not
434 diagnostic discrimination or disease monitoring. Fourth, the retroactive correction adds one-stride latency
435 and prevents real-time estimation, although it does not affect offline applications. Fifth, single-session
436 data leave test-retest reliability and responsiveness to clinical change untested.

437 5 CONCLUSIONS

438 A calibration-free heel-mounted IMU pipeline estimated stride length in older adults to within 3.9 cm
439 RMSE (ICC 0.976) across single-task and graded dual-task walking. Per-condition RMSE ranged from
440 3.7 to 4.0 cm, and error was low in both MoCA-defined groups. Stride-length variability was less
441 accurate than mean stride length. These findings establish criterion validity for successfully synchronised,
442 supervised straight walkway passes. They do not establish diagnostic utility, longitudinal responsiveness,
443 or everyday-living accuracy. Independent validation should test other clinical cohorts, repeated sessions,
444 and unsupervised walking with turns, stops, varied footwear, and uneven surfaces.

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450 **COMPETING INTERESTS**

451 The authors declare that they have no competing interests.

452 **AUTHOR CONTRIBUTIONS**

453 **Pawel Kudzia** conceived the algorithm and analysis approach, designed the data-processing pipeline,
454 performed the data analysis, prepared figures and tables, drafted the manuscript, and approved the final
455 draft.

456 **Markus von Hacht** contributed to data collection, reviewed and edited drafts of the article, and approved
457 the final draft.

458 **Maryam Sheikhi** contributed to data collection and analysis, reviewed and edited drafts of the article,
459 and approved the final draft.

460 **Drew Commandeur** contributed to development of the data-collection and synchronisation methods,
461 reviewed and edited drafts of the article, and approved the final draft.

462 **Marc Klimstra** contributed to development of the data-collection and synchronisation methods, reviewed
463 and edited drafts of the article, and approved the final draft.

464 **Sandra Hundza** conceived and designed the experiments, supervised data collection, supervised the
465 analysis, reviewed and edited drafts of the article, and approved the final draft.

466 **HUMAN ETHICS**

467 The University of Victoria Human Research Ethics Board approved this study (protocol H22-00451). All
468 participants provided written informed consent.

469 DATA AVAILABILITY

470 We will deposit the de-identified stride-level data and the analysis code that support the findings of this
471 study in a public repository (Zenodo) and release them under an open licence upon publication. We will
472 additionally host the analysis code on GitHub, and we will provide a persistent DOI and the repository
473 URL at that time.

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